

PKI G3 Certificate Request (CSR)

Document: PEPPOL Production PKI (G3) Certificate Signing Request **Service Provider:** AL MERAK TAX CONSULTANT L.L.C (E-Numerak) **PEPPOL Seat ID (test):** PAE001147 **Date:** 2026-06-20

1. Purpose

To obtain a **production** PEPPOL Access Point certificate (PKI G3), OpenPEPPOL's Certificate Authority (operated by DigiCert) must receive a **Certificate Signing Request (CSR)** generated from a private key that **never leaves our server**. The CA signs the CSR and returns the public certificate; the private key stays with us.

We currently hold only a **test** certificate (Issuer: PEPPOL ACCESS POINT TEST CA - G3 , OU=PEPPOL TEST AP). UAE accreditation requires re-passing all testbed suites with a **production** certificate (Issuer: PEPPOL ACCESS POINT CA - G3).

2. Certificate subject (Distinguished Name)

OpenPEPPOL **assigns** the Common Name (the Seat ID). For production it is the PAE/POP seat issued to us during onboarding. The remaining fields are fixed by the PEPPOL CA Certificate Policy:

Field	Value	Notes
CN (Common Name)	PAE001147 (production seat — confirm with OpenPEPPOL)	Assigned by OpenPEPPOL
O (Organisation)	AL MERAK TAX CONSULTANT L.L.C	Legal entity name
OU (Org Unit)	PEPPOL AP	Production value (test uses PEPPOL TEST AP)
C (Country)	AE	ISO 3166 country code

Key requirements (PEPPOL CP G3): - Algorithm: **RSA 2048-bit** (minimum) — we use 2048. - Signature: **SHA-256**. - Key usage: Digital Signature + Key Encipherment (set by the CA, not the CSR).

3. How the CSR is generated (OpenSSL)

Run the helper script (added to the repo):

```
bash scripts/generate_peppol_csr.sh PAE001147 "AL MERAK TAX CONSULTANT L.L.C"
```

This produces, in certs/production/ : - peppol-ap-prod.key — **2048-bit RSA private key (KEEP SECRET, never email/commit)** - peppol-ap-prod.csr — the CSR to upload to OpenPEPPOL

Equivalent raw OpenSSL commands (what the script does):

```
# 1. Generate a 2048-bit RSA private key
openssl genrsa -out peppol-ap-prod.key 2048

# 2. Generate the CSR with the PEPPOL subject DN
openssl req -new -sha256 \
  -key peppol-ap-prod.key \
  -out peppol-ap-prod.csr \
  -subj "/C=AE/O=AL MERAK TAX CONSULTANT L.L.C/OU=PEPPOL AP/CN=PAE001147"

# 3. Verify the CSR before submitting
openssl req -in peppol-ap-prod.csr -noout -text -verify
```

4. Submission flow

1. Generate key + CSR with the script above (**on the production server**, so the private key never travels).
2. Log into the **OpenPEPPOL onboarding / Service Metadata portal** (access given during enrolment by our PEPPOL Authority — for UAE this is coordinated with the FTA / Ministry of Finance).
3. Upload `peppol-ap-prod.csr`.
4. OpenPEPPOL CA issues the production certificate (`.cer` / `.pem`).
5. Assemble the full chain → see [02-certificate-chain-trust-proof.md](#).
6. Install in the keystore and re-run all 4 PINT-AE suites.

5. Security handling of the private key

- The `.key` file is generated with `umask 077` (owner-read-only).
- It is stored under `certs/production/` which is **git-ignored** and excluded from backups that leave UAE.
- For runtime we package key + issued cert into a password-protected PKCS#12 (`.pfx`) — same format as the current `certs/ap-test.p12.pfx`.
- The keystore password is supplied via environment variable (`PEPPOL_KESTORE_PASSWORD`), never hard-coded.

6. Checklist

- ☐ Confirm the **production Seat ID (CN)** with OpenPEPPOL.
- ☐ Generate key + CSR **on the production host**.
- ☐ Verify CSR subject (`openssl req -verify`).
- ☐ Upload CSR to OpenPEPPOL portal.
- ☐ Receive issued certificate.
- ☐ Build chain & PKCS#12 (next doc).

- ☐ Re-run testbed suites with production cert.